Year 13 Further Maths - Matrices

Similar questions - Determinants, singular matrices, scale factors

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathbf{A} = \begin{pmatrix} a & 3 & 0 \\ 0 & a & 1 \\ 2 & 5 & 1 \end{pmatrix}$ where a is a real constant.

(a) Show that $\det \mathbf{A} = a^2 - 5a + 6$. **[2]**

(b) Find the values of a for which $\mathbf{A}$ is singular. **[2]**

(c) Consider the equations $ax + 3y = 6$, $ay + z = 5$, $2x + 5y + z = 11$. For each value of a found in (b), determine whether the equations have a unique solution, infinitely many solutions or no solution. **[6]**

2 $\mathbf{M} = \begin{pmatrix} 2 & -0.6 \\ 1.5 & 0.8 \end{pmatrix}$ represents a stretch parallel to one of the coordinate axes followed by a rotation about the origin.

(a) Find $\det \mathbf{M}$. **[1]**

(b) By considering determinants, state the scale factor of the stretch. **[2]**

(c) Explain whether the stretch is parallel to the x-axis or the y-axis. **[2]**

(d) Find the angle of rotation. **[2]**

3 A transformation T is represented by $\mathbf{N} = \begin{pmatrix} a & 2 & 0 \\ 0 & 1 & 3 \\ 1 & 0 & 2 \end{pmatrix}$ where a is a constant.

(a) Find $\mathbf{N}^2$ in terms of a. **[3]**

(b) Find $\det \mathbf{N}$ in terms of a. **[2]**

Given that $a = 13$ to the nearest integer, a shape S_1 has volume 11.6 (1 decimal place) and is mapped to S_2 by T. A student claims the volume of S_2 is less than 400.

(c) Comment on the student's claim. **[3]**

Question 1

(a) $\det \mathbf{A} = a(a \cdot 1 - 5) - 3(0 \cdot 1 - 1 \cdot 2) = a^2 - 5a + 6$

(b) $(a-2)(a-3) = 0$, so $a = 2$ or $a = 3$

(c) Take R_1, R_2, R_3 to be the three equations. For $a = 2$: $R_3 - R_1 - R_2$ gives $0 = 11 - 6 - 5 = 0$, consistent, so the equations are dependent: infinitely many solutions. For $a = 3$: $R_3 - \frac{2}{3}R_1 - R_2$ gives $0 = 11 - 4 - 5 = 2$, a contradiction, so there is no solution

Question 2

(a) $\det \mathbf{M} = 1.6 + 0.9 = 2.5$

(b) Rotation has determinant 1, so the stretch has scale factor 2.5

(c) $\mathbf{M} = \mathbf{RS}$ with $\mathbf{S} = \text{diag}(k, 1)$ or $\text{diag}(1, k)$. The first column of $\mathbf{M}$ has length $\sqrt{4 + 2.25} = 2.5$ and the second column has length 1, so the x-direction is stretched: parallel to the x-axis

(d) $\mathbf{R} = \mathbf{MS}^{-1} = \begin{pmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{pmatrix}$, so $\cos\theta = 0.8$, $\sin\theta = 0.6$ and $\theta = 36.9°$ anticlockwise

Question 3

(a) $\mathbf{N}^2 = \begin{pmatrix} a^2 & 2a+2 & 6 \\ 3 & 1 & 9 \\ a+2 & 2 & 4 \end{pmatrix}$

(b) $\det \mathbf{N} = a(2-0) - 2(0-3) = 2a + 6$

(c) For $12.5 \leqslant a < 13.5$, $\det \mathbf{N}$ lies between 31 and 33. The volume of S_1 lies in $[11.55, 11.65)$, so the volume of S_2 lies between $31 \times 11.55 = 358.05$ and $33 \times 11.65 = 384.45$, which is below 400. The student's claim is correct